UNIT - 03

"HOSPITAL PATIENT MANEGEMENT SYSTEM..."

★ **Problem Statement :** Develop a hospital patient manegement system using a ~~to~~ doubly linked list. a stack and an array Store patient details such as patient ID name , age , disese and alloted bed number in a doubly linked list . The system should . allow admitting new patients, discharging patients searching for a patient and displaying all patient records in foreword and backword order . hospital beds are maneged using an array for bed allocation and discharged patients are saved in a stack so that the last discharge can be undone.

★ **Objective :** The objective of this project is to develop a hospital patient manegement system in C++ using structures without classes patient details such as

ID, name, age disease and allocated bed number are stored in a doubly linked List which allows easy insertion, deletion searching and traversal in both forword and backword direction. Hospital beds are maneged using an array so the system can allot a free bed to each new patient are stored in a stack which keeps the discharge history and lets the last discharge be undone. through this program we apply linked list stack and array concepts to solve real-world hospital problem efficiently.

★ Algorithm :-

Step 1: Start. Create an empty doubly linked list an array of 10 free beds and an empty Stack.

Step 2: Display the menu : admit, discharge search display forword display backword bed status cmdo discharge history exit.

Step 3: Read the choice and call the matching function.

Step 4: Repeat steps 2 and 3 and 3 untill choice = 0

Step 5: Find the first free bed in the array if none is free display "No beds available" and stop.

Step 6: Read patient ID if the ID already exists in the list stop.

Step 7: Read name age and diseses

Step 8: Create a new node and insert it at the end of the list (update tail and prev / next links)

Step 9: Mark the bed as occupied and increse count.

Step 10: Read patient ID and search the list from head if not found display "patient not found" and stop

Step 11: push the patient record into stack (top=+1

Step 12: free the patient bed in the array

Step 13: Delete the node by linking its previous and next nodes (update head or tail if needed)

Step 14: Decrese count

Step 15: Read the patient ID or more.

Step 16: Traverse the list from head uying next untill NULL.

Step 17: If a node matches display the patient details otherwise display "petient not found".

(4)

Step 18 : Forward : Start at head and print each node using next until NULL

Step 19 : Backward : Start at tail and print each node using prev until NULL.

Step 20 : If the desk is empty, display a message and stop.

Step 21 : pop the top record (top = top-1)

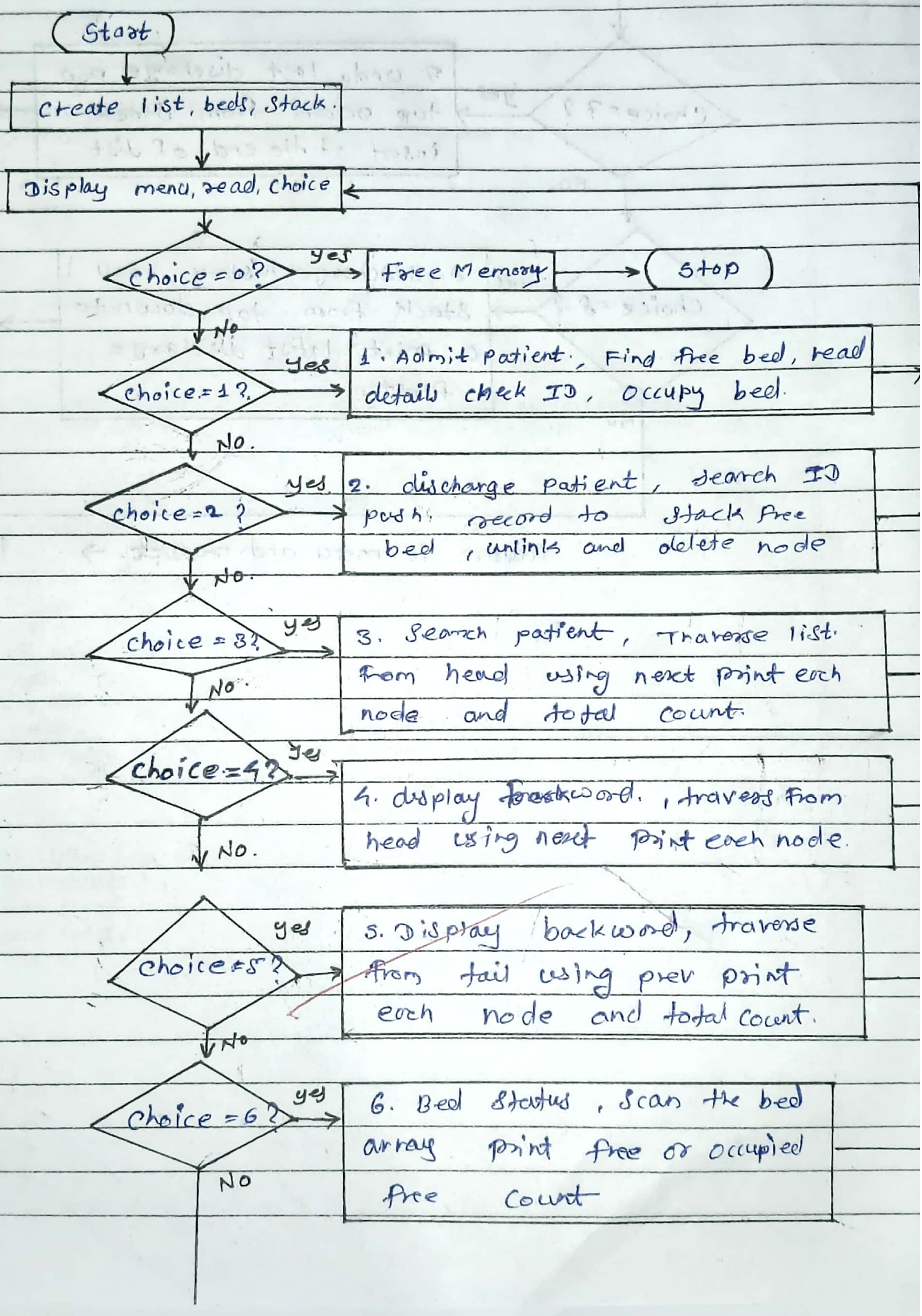
Step 22 : Insert the patient at the end of the list and mark a free bed as occupied.

Step 23 : Bed status : traverse the bed array and print each bed as free or occupied

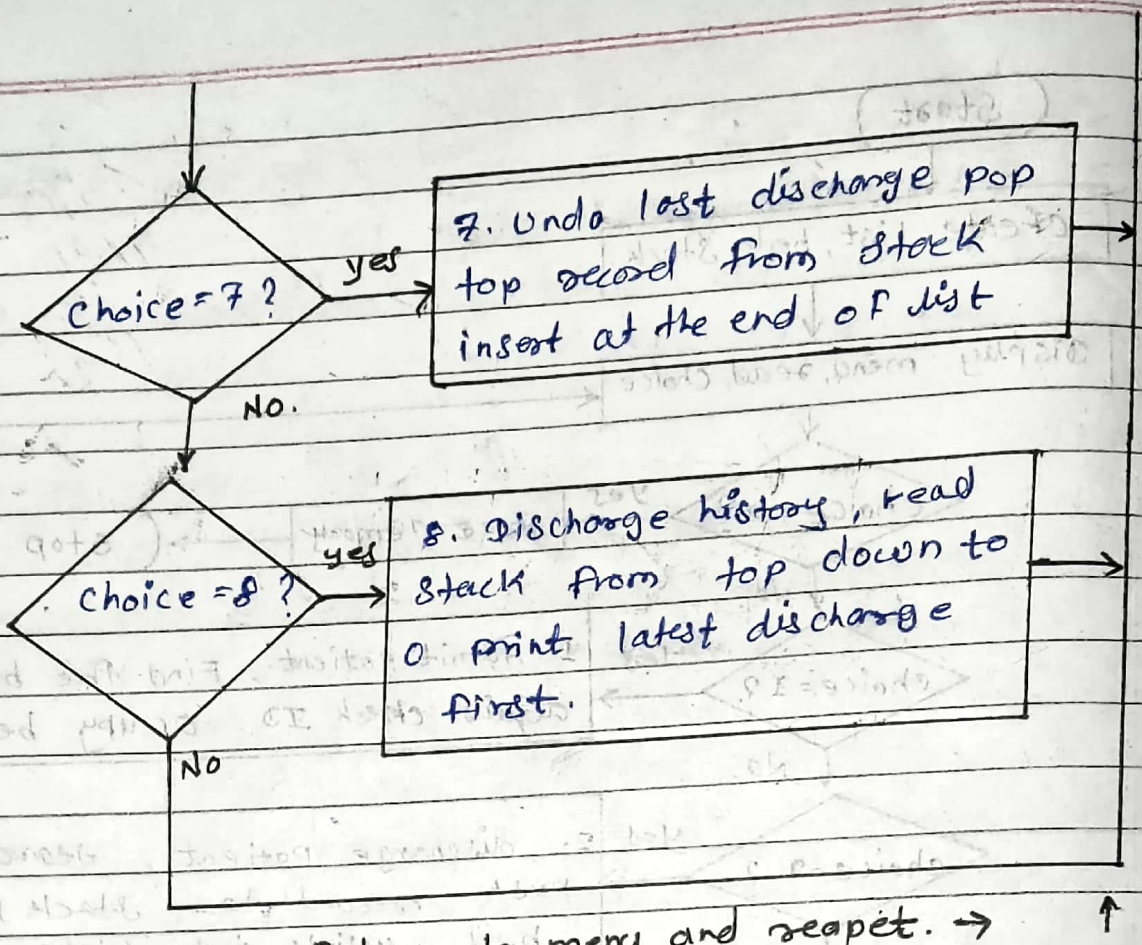
Step 24 : History : print the stack records from top to 0 (latest discharge first)

Step 25 : Stop.

★ Flowchart :-


Start
Create list, beds, Stack.
Display menu, read, choice
Choice = 0?
yes
Free Memory
Stop
No
Choice = 1?
yes
1. Admit Patient., Find free bed, read details check ID, Occupy bed.
No.
choice = 2?
yes
2. discharge patient, search ID push record to Stack free bed, unlink and delete node
No.
Choice = 3?
yes
3. Search patient, Traverse list. from head using next print each node and total count.
No.
Choice = 4?
yes
4. display backword., travers from head using next print each node.
No.
Choice = 5?
yes
5. Display backword, traverse from tail using prev print each node and total count.
No
Choice = 6?
yes
6. Bed Status, Scan the bed array print free or occupied free count
No

(6)


Choice = 7 ?
yes
7. Undo last discharge pop top record from stack insert at the end of list
No.
Choice = 8 ?
yes
8. Discharge history, read stack from top down to 0 print latest discharge first.
No
Return to menu and reapet. →